

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 7: Ratio and Proportion

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Find the duplicate ratio and the sub-duplicate ratio of 9 : 16.
2. Find the compound ratio of 3 : 4, 8 : 9 and 3 : 2.
3. If $x : y = 4 : 7$, find $(3x + 2y) : (5x + y)$.
4. Find the fourth proportional to 4, 7 and 12.
5. Find the mean proportional between 8 and 18.
6. Find the third proportional to 9 and 12.
7. What number must be subtracted from each of 7, 17 and 47 so that the results are in continued proportion?
8. If $\frac{5x + 2y}{5x - 2y} = \frac{4}{3}$, find $x : y$.
9. Solve $\frac{\sqrt{x+7} + \sqrt{x-8}}{\sqrt{x+7} - \sqrt{x-8}} = \frac{5}{3}$.
10. If $\frac{a}{b} = \frac{c}{d}$, prove that $\frac{a+c}{b+d} = \frac{a}{b}$.

Answer Key

1. $81 : 256$ and $3 : 4$

2. $72 : 72 = 1 : 1$

3. $26 : 27$

4. 21

5. 12

6. 16

7. $(17 - x)^2 = (7 - x)(47 - x)$ gives $x = 2$

8. $\frac{5x}{2y} = 7$, so $x : y = 14 : 5$

9. $\frac{\sqrt{x+7}}{\sqrt{x-8}} = 4$, so $x + 7 = 16(x - 8)$ and $x = 9$

10. With $a = bk, c = dk$: $\text{LHS} = \frac{k(b+d)}{b+d} = k = \text{RHS}$